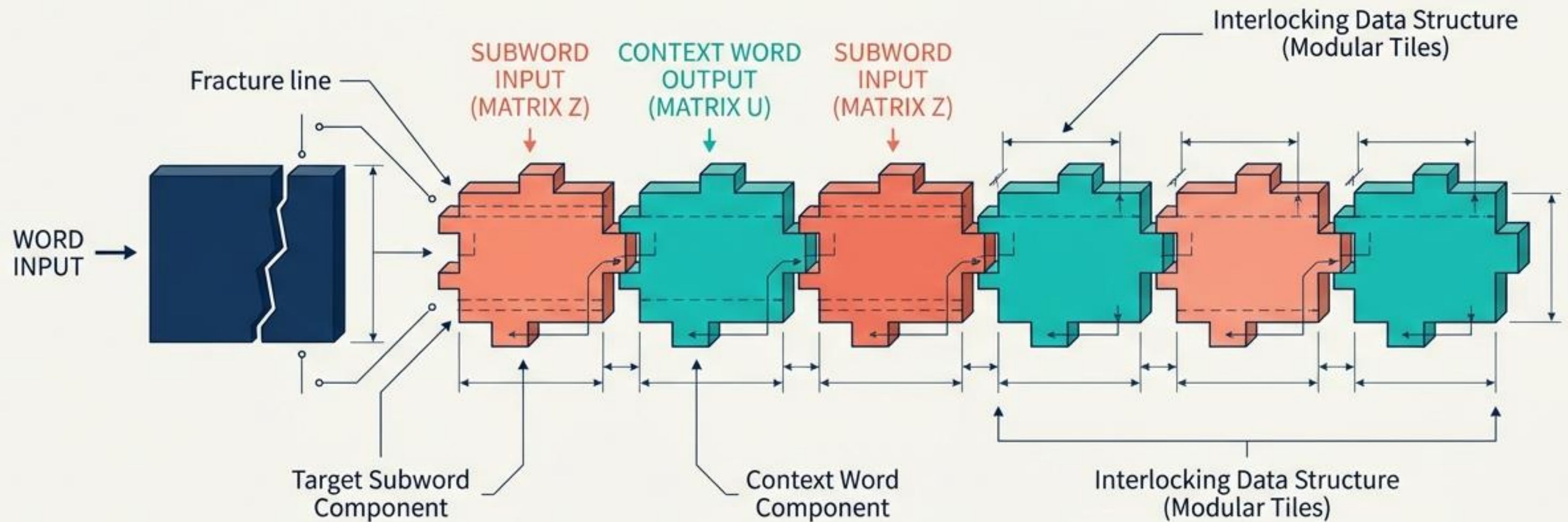


Deconstructing FastText

The Architecture of Subword Tokenisation and Semantic Embedding

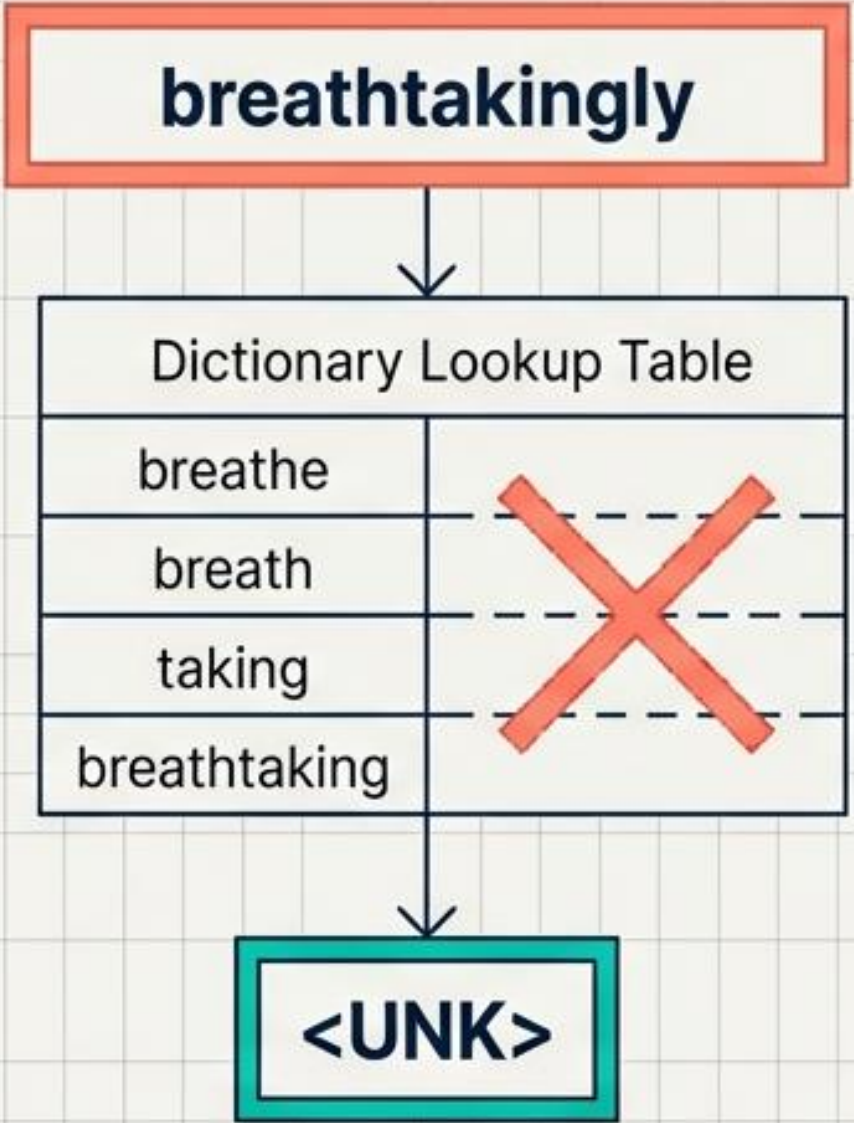
Based on research by Dr Partha Majumdar (ORC-ID: 0009-0002-7375-8034)



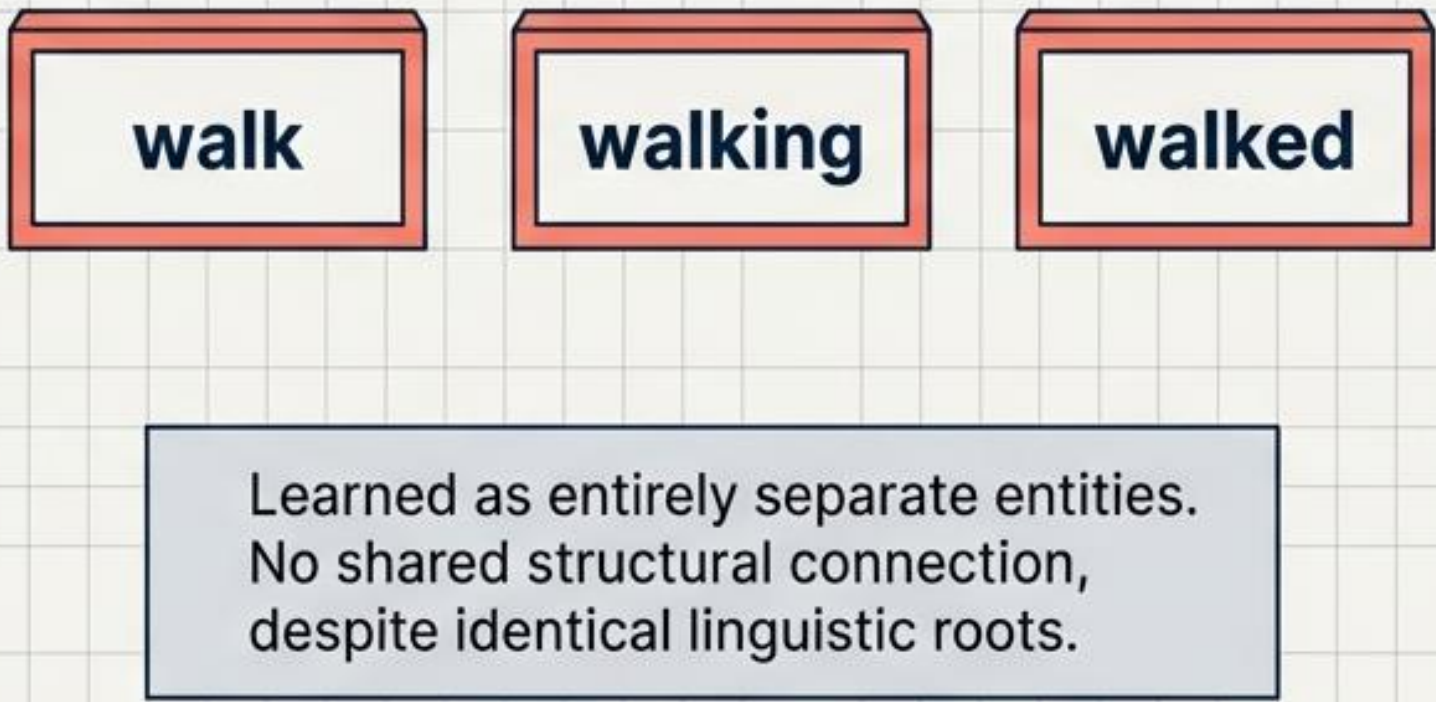
The Atomic Limitation in Early NLP

Models like Word2Vec and GloVe share a fundamental syntactic blind spot: they treat every word as an indivisible, atomic unit of meaning.

The Out-Of-Vocabulary Failure



The Morphological Blindness



The Sparsity Crisis in Complex Morphology

Treating words as atomic units causes severe vocabulary bloat in languages that rely heavily on inflexion, derivation, or compounding.



Hausaufgabe



Haus



Aufgabe



schneller



schnell

Without a mechanism to look inside the word, the model requires massive vocabularies and exorbitant training data to capture simple grammatical variations.

The Subword Paradigm

FastText disrupts the atomic assumption. A word is no longer a fixed symbol, but a compositional aggregation of overlapping character sequences.

Anatomy of a Word

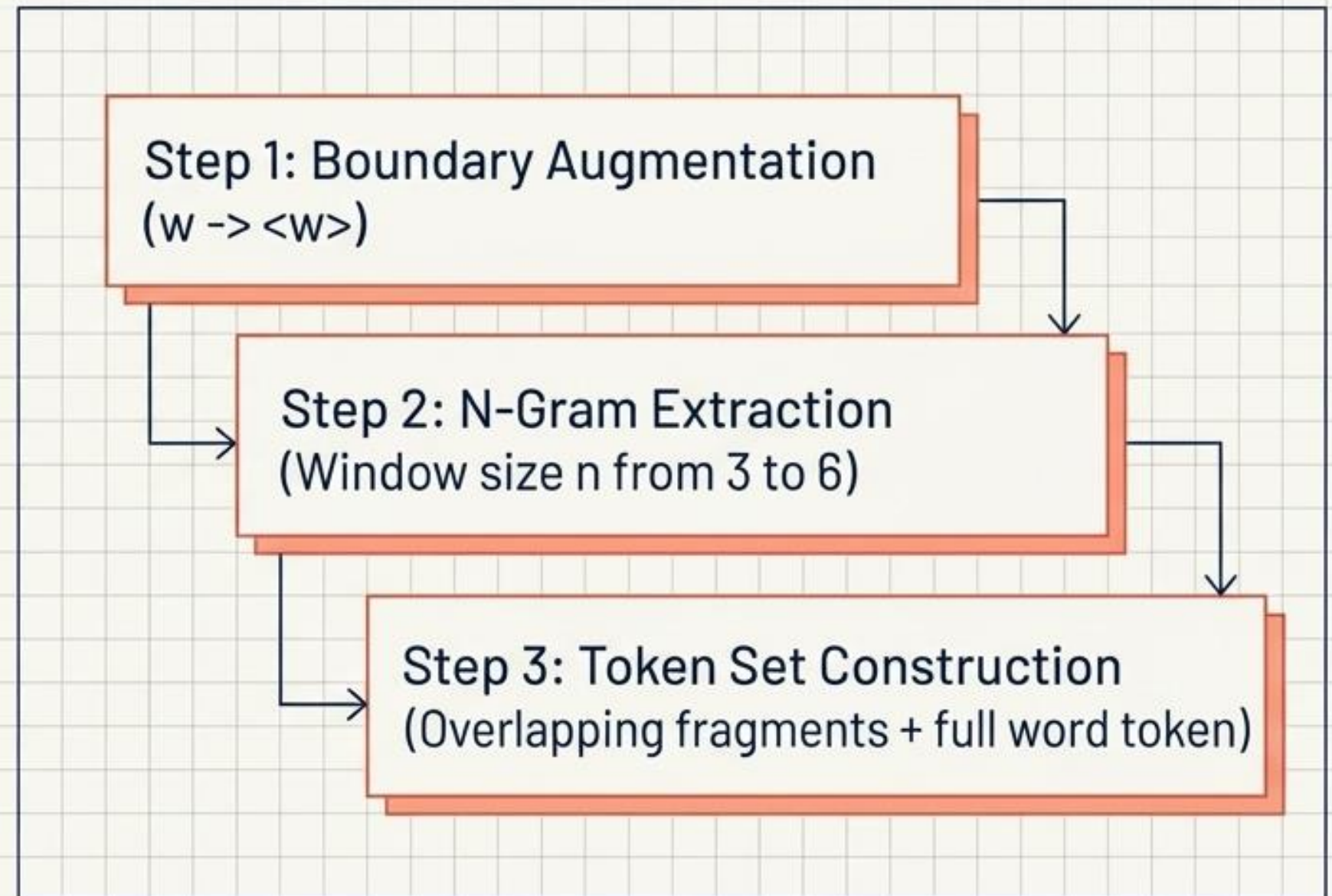
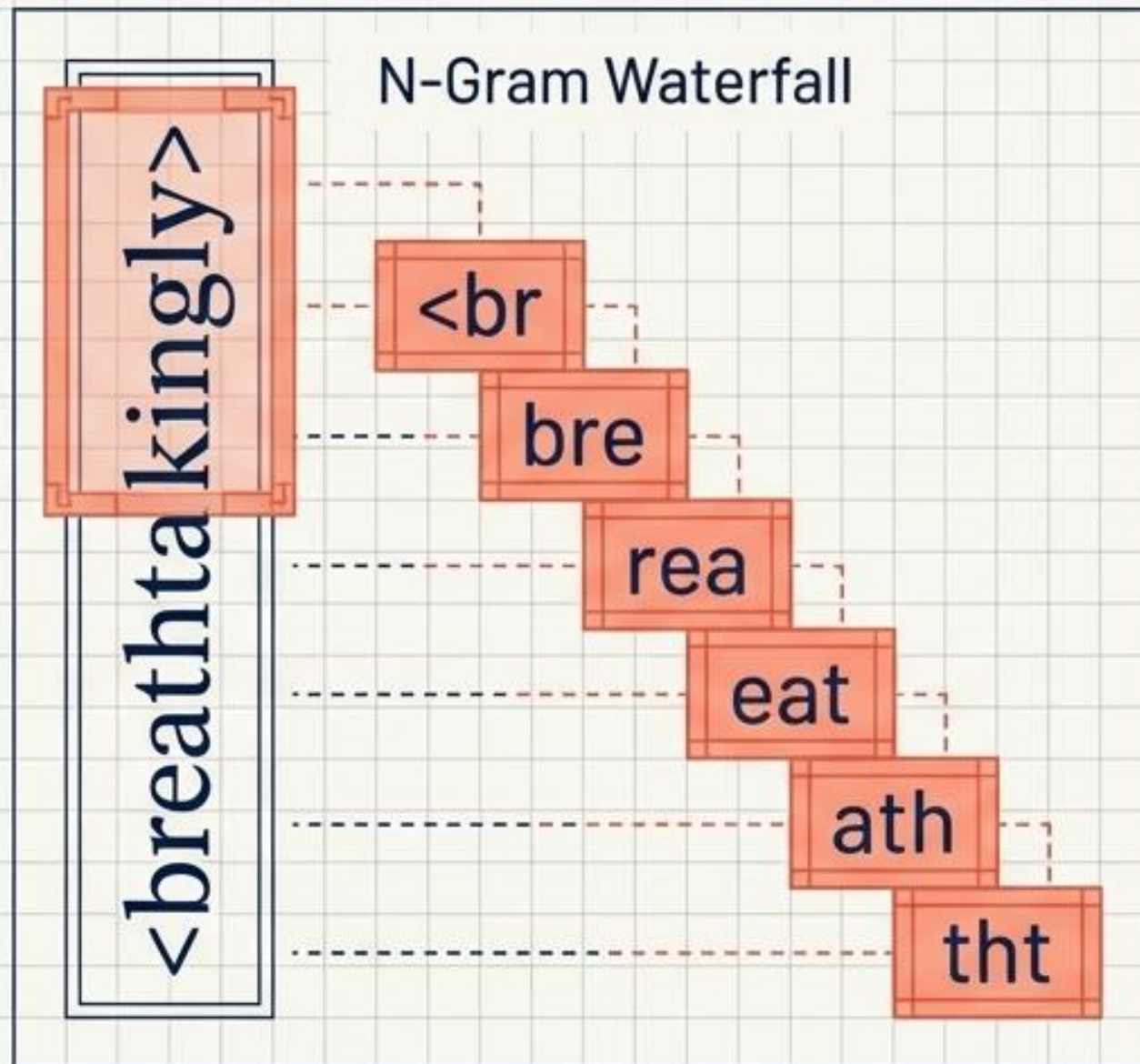
<fast>



Boundary markers < and > preserve crucial positional information, distinguishing a prefix from a middle syllable.

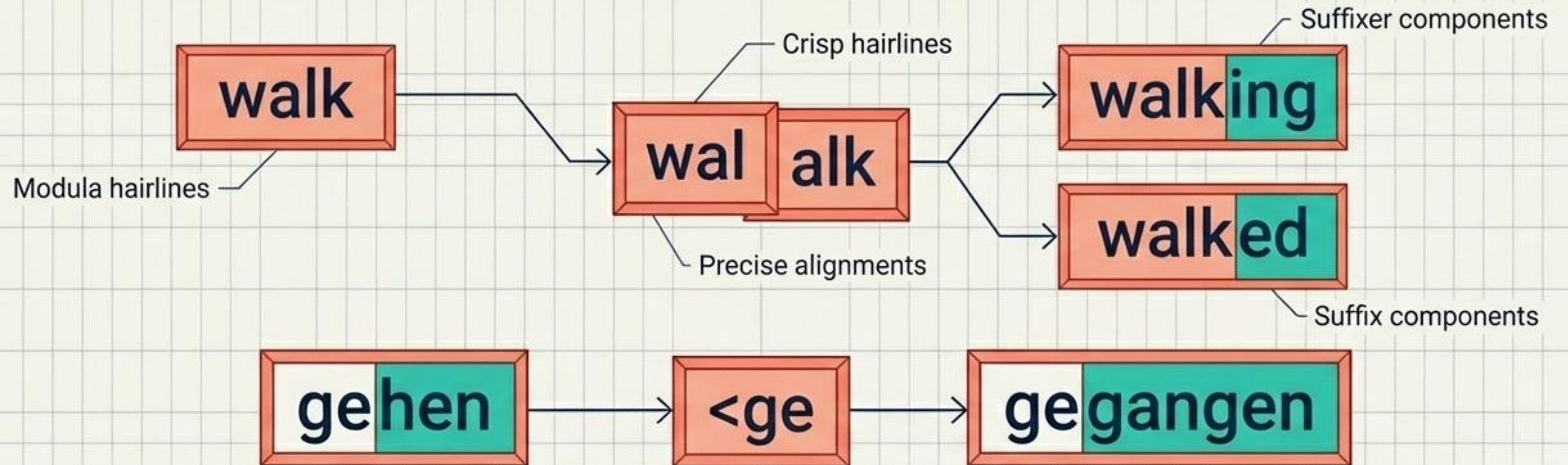
The Sliding Window Mechanism

The mapping of a word to its subcomponents is not learned; it is an algorithmically fixed, deterministic procedure ensuring consistent decomposition.



Sharing Statistical Strength

By breaking words into reusable components, updates to one word automatically propagate to morphological relatives.



This deterministic decomposition elegantly solves the Out-Of-Vocabulary problem. A novel word simply triggers the assembly of familiar, pre-learned subcomponents.

The NLP Architecture Diagnostic

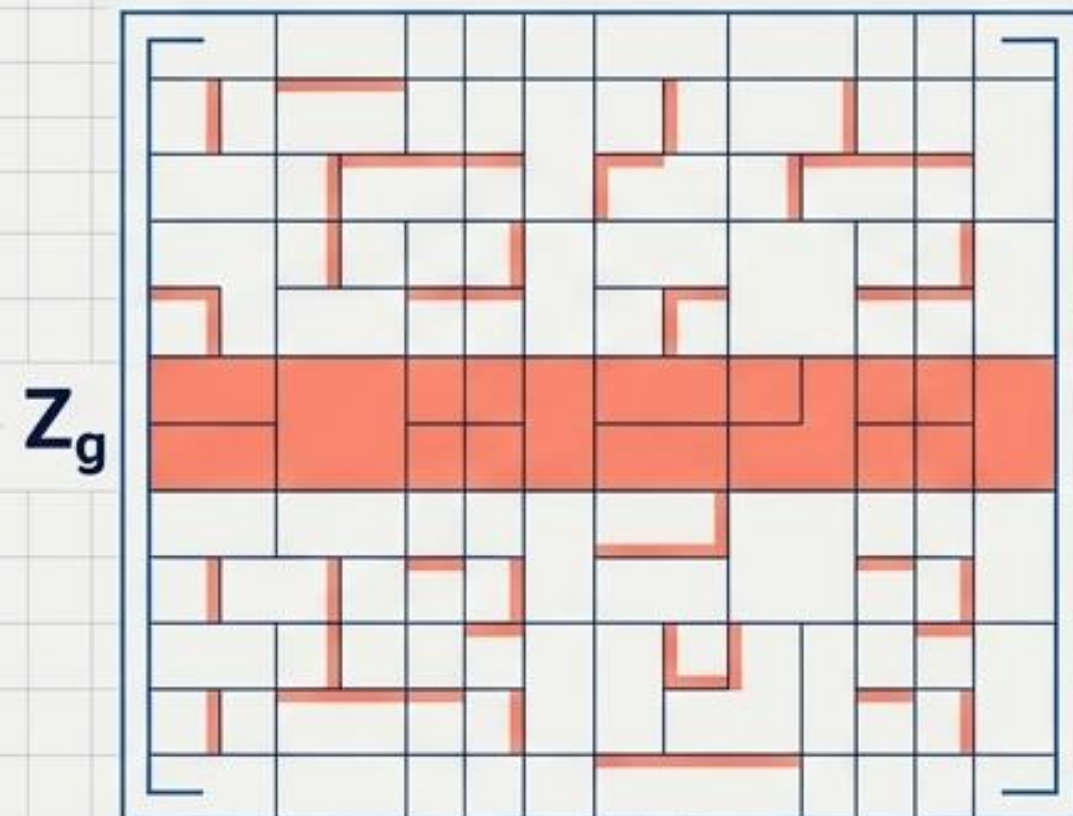
A structural comparison of word representation methodologies.

Dimension	Atomic (Word2Vec/GloVe)	Compositional (FastText)
Definition of a Word	Indivisible symbol	Aggregation of subwords
OOV Handling	Fails, returns <UNK>	Generates inferred embeddings
Morphological Awareness	Blind to internal structure	Explicitly captures shared affixes
Data Sparsity	High (needs massive dictionary)	Low (reuses n-gram features)
Memory Footprint	Scales with exact vocabulary	Scales with distinct character n-grams

The Mechanics of Meaning: Matrices Z and U

FastText translates structural representation into numerical vectors through two distinct embedding matrices.

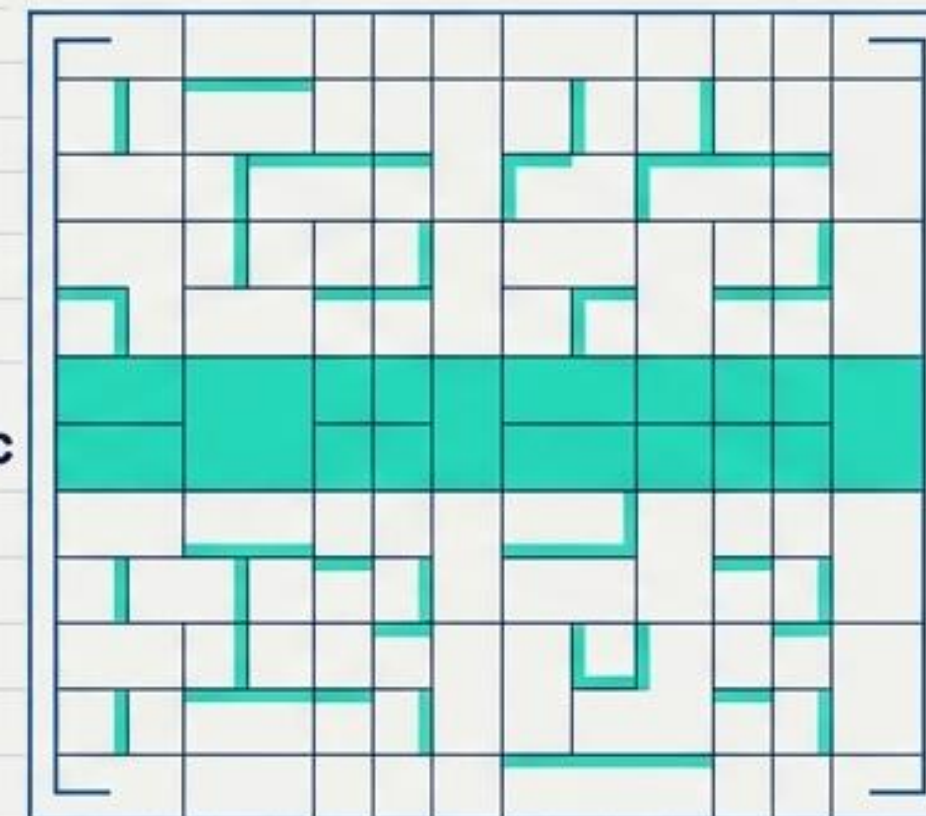
Matrix Z: The Subword Embedding Matrix



<pl

U_c

Matrix U: The Output Word Embedding Matrix



games

$$h_w = \frac{1}{|G_w|} \sum (Z_g)$$

A word's embedding (h_w) is the direct mathematical average of its constituent subword vectors from Matrix Z.

The Forward Pass: Theory to Code

Evaluating the semantic compatibility between an input word and a context word space.

The Mathematics	PyTorch Execution
1. Aggregate target subwords: $\mathbf{h}_w$	<code>target_word_vector = target_subword_vectors.mean(dim=0)</code>
2. Retrieve context vector: $\mathbf{u}_c$	<code>context_word_vector = output_embedding_matrix[context_word_index]</code>
3. Compute compatibility: $s(\mathbf{w}, \mathbf{c}) = \mathbf{h}_w * \mathbf{u}_c$	<code>score = torch.dot(target_word_vector, context_word_vector)</code>
4. Probability: $P(\mathbf{c} \mathbf{w}) = \text{Sigmoid}(\mathbf{h}_w * \mathbf{u}_c)$	<code>probability = torch.sigmoid(score)</code>

Generating the Training Dataset

Neural language models learn from contextual relationships. FastText dynamically **harvests positive target-context pairs** using a **sliding window**.



Algorithmic Harvest

Pair 1: (playing, are) → Label: 1

Pair 2: (playing, games) → Label: 1

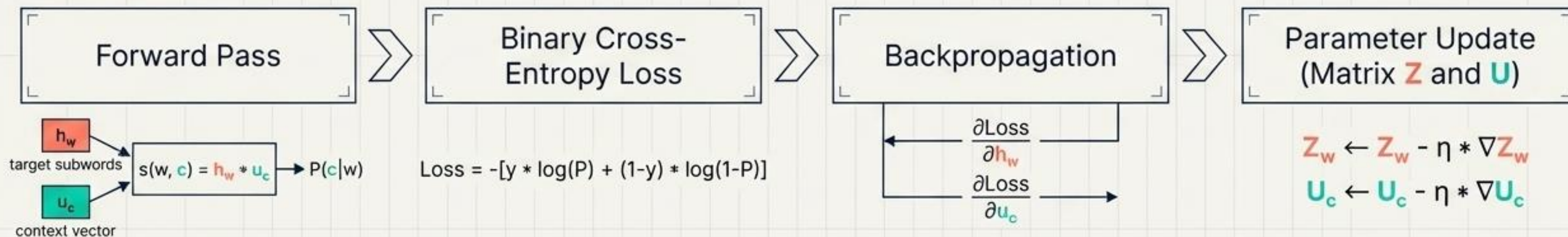
Negative Sample: (playing, children) → Label: 0

← Stochastic noise injected to define the boundaries of the semantic space.

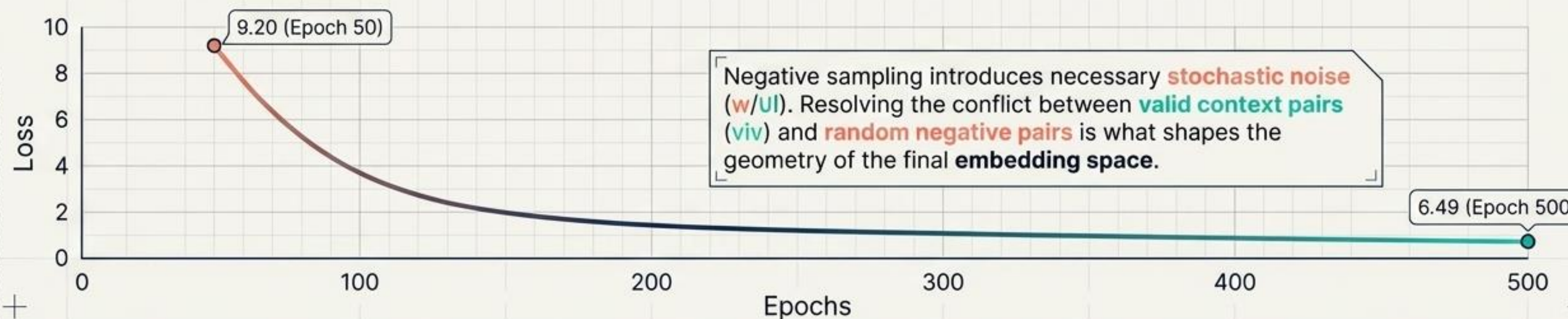
The Neural Network Training Loop

Iterating over the complete generated dataset to shape a discriminative embedding space.

Training Loop Flowchart



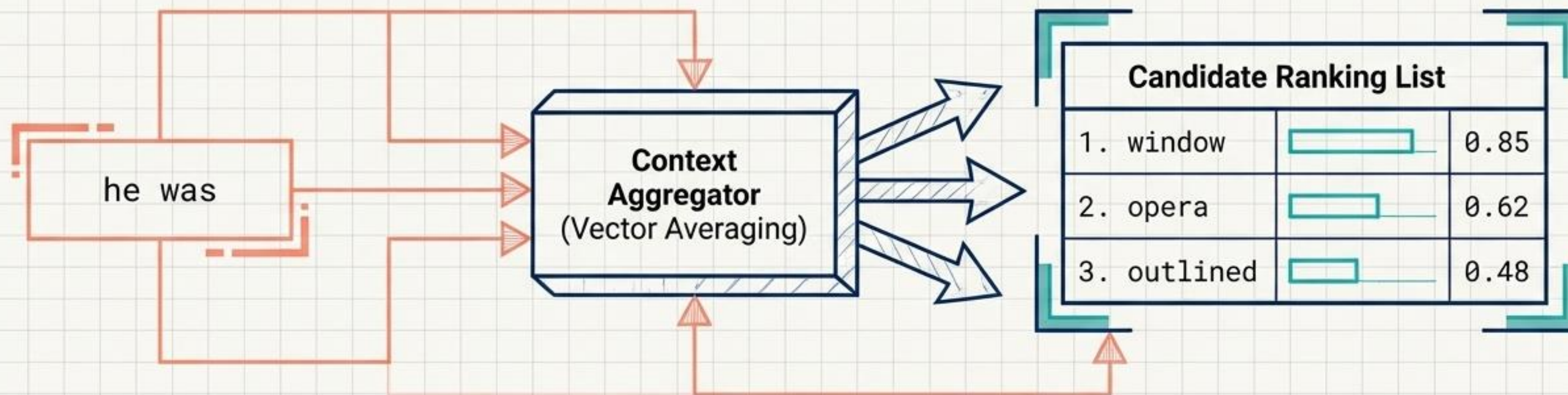
Training Loss Over Epochs



Application I: Next-Word Prediction

Transforming the trained FastText embedding space into a practical inference engine using a **12,000-token** corpus from **Arthur Conan Doyle**.

Inference Engine Schematic



Because FastText builds from character n-grams, it gracefully handles rare or inflected vocabulary inherent in 19th-century prose.

Semantic Compatibility in Action

The model ranks candidate vocabulary words by their mathematical similarity to the input context vector.

Terminal Output

Input: 'sherlock'
→
Predicted:
state, excuse,
worker

Terminal Output

Input: 'he was'
→
Predicted:
window, opera,
outlined

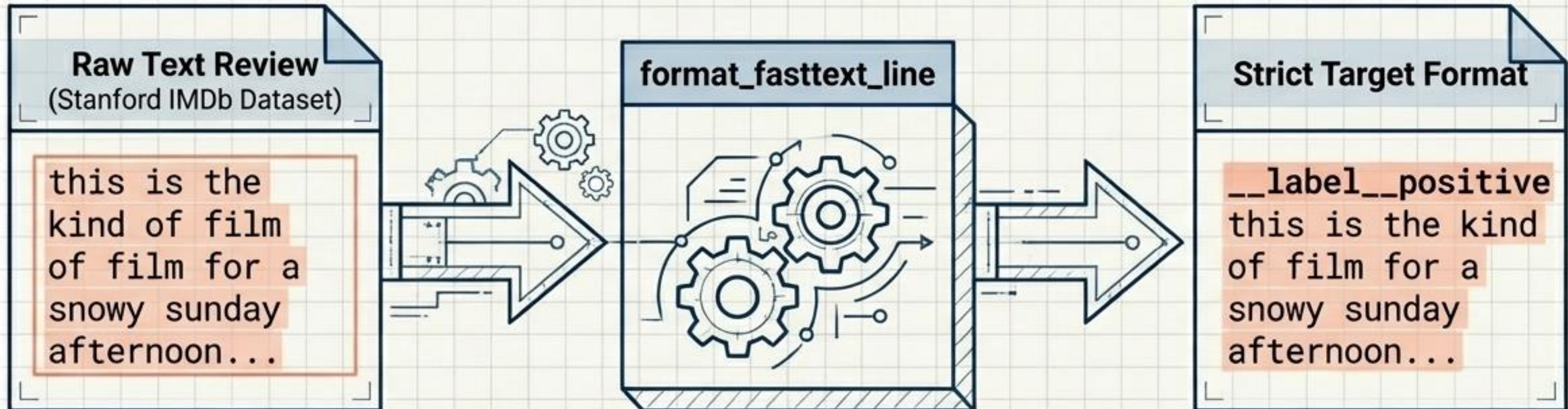
Terminal Output

Input: 'i could'
→
Predicted:
statement, setter,
thought

The model relies entirely on structural subword mathematics to infer local syntactic compatibility, generating logically sound continuations.

Application II: Sentiment Analysis

Moving from manual PyTorch matrices to production-style workflows using the official fastText Python library.



```
fasttext.train_supervised(input=train_file, lr=0.5, wordNgrams=2, dim=100)
```


Prediction vs. Reality

Evaluating the supervised FastText model across 25,000 unseen test reviews.

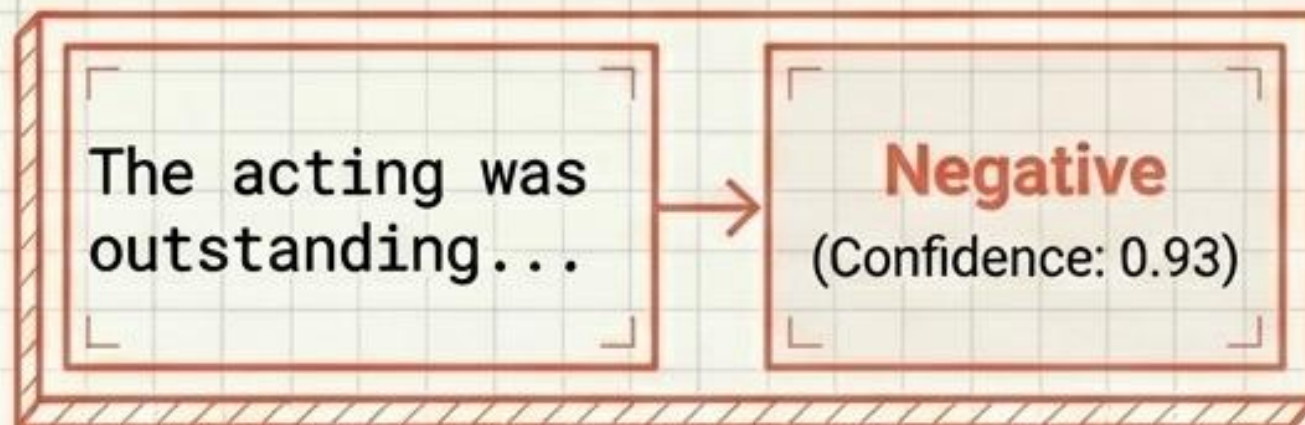
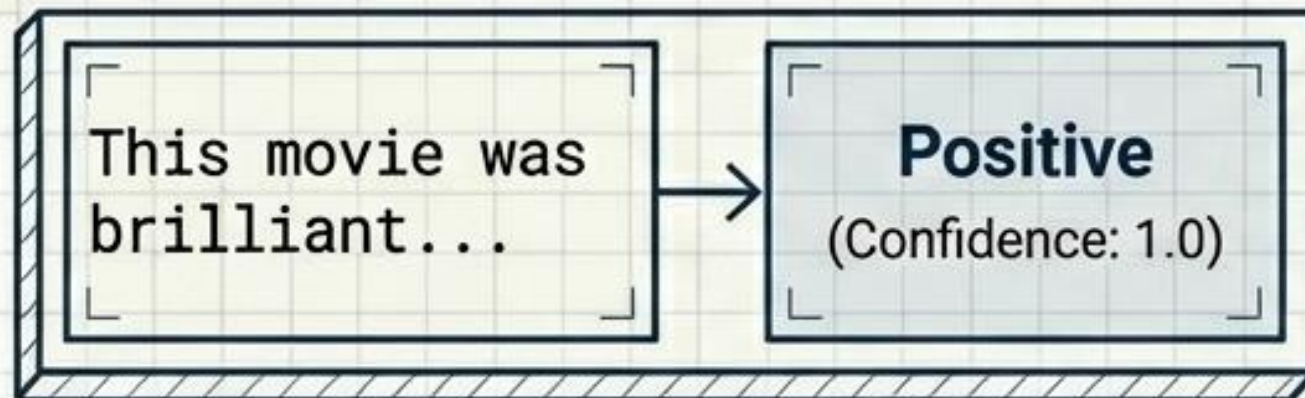
		Predicted	
		Positive	Negative
Actual	Positive	10,966	1,329
	Negative	1,534	11,171

Strong diagonal dominance proves the model accurately captures polarity, with balanced accuracy across both positive and negative semantic expressions.

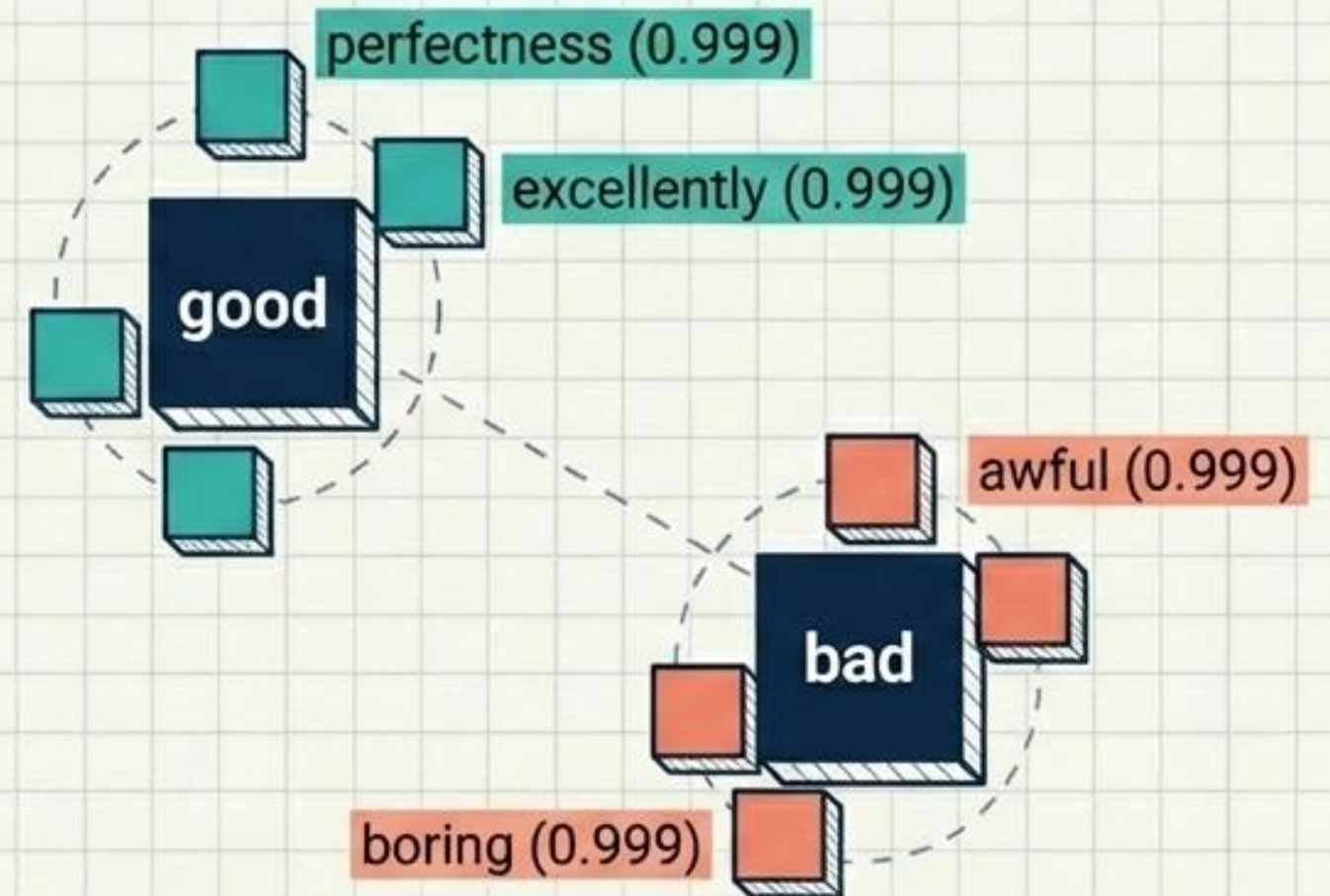
Inspecting Semantic Geometry

The model achieves saturated confidence on polarized language, but reveals edge cases in nuanced expression.

Confidence Probe

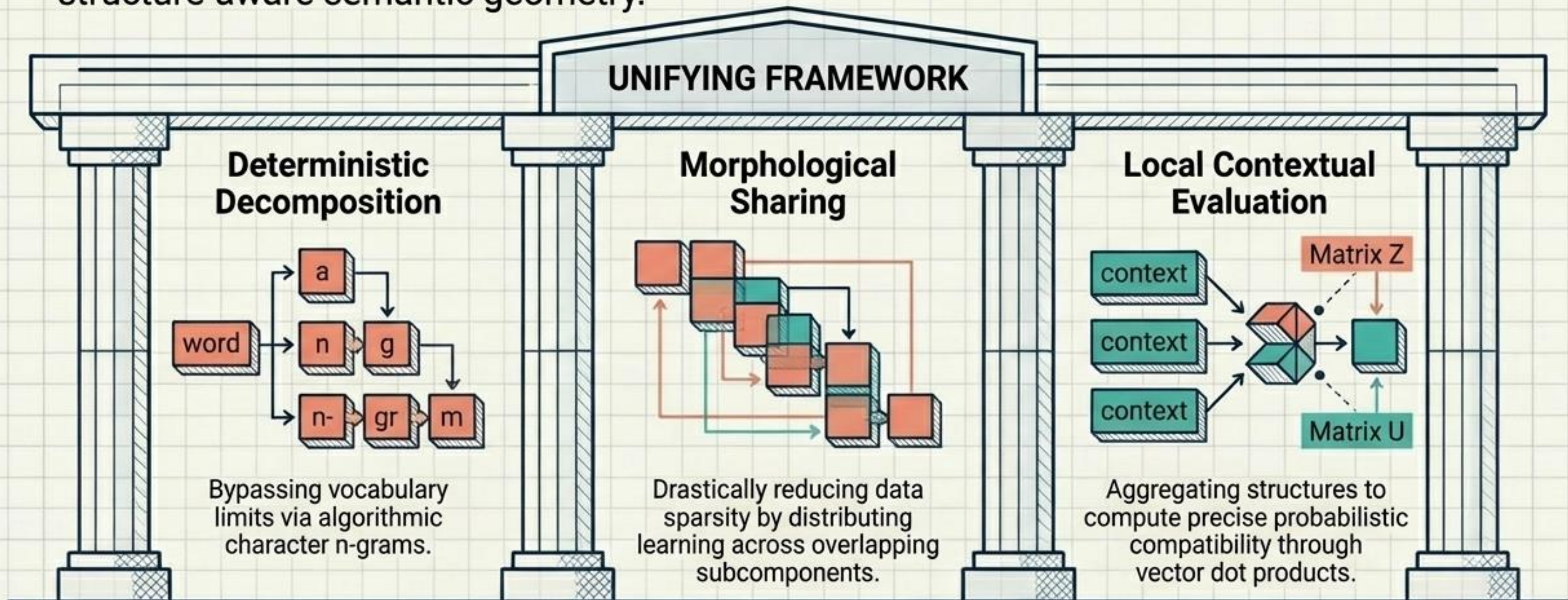


Nearest Neighbours Map



Synthesis: From Subwords to Semantics

FastText represents a conceptual shift from isolated word-level representation to structure-aware semantic geometry.

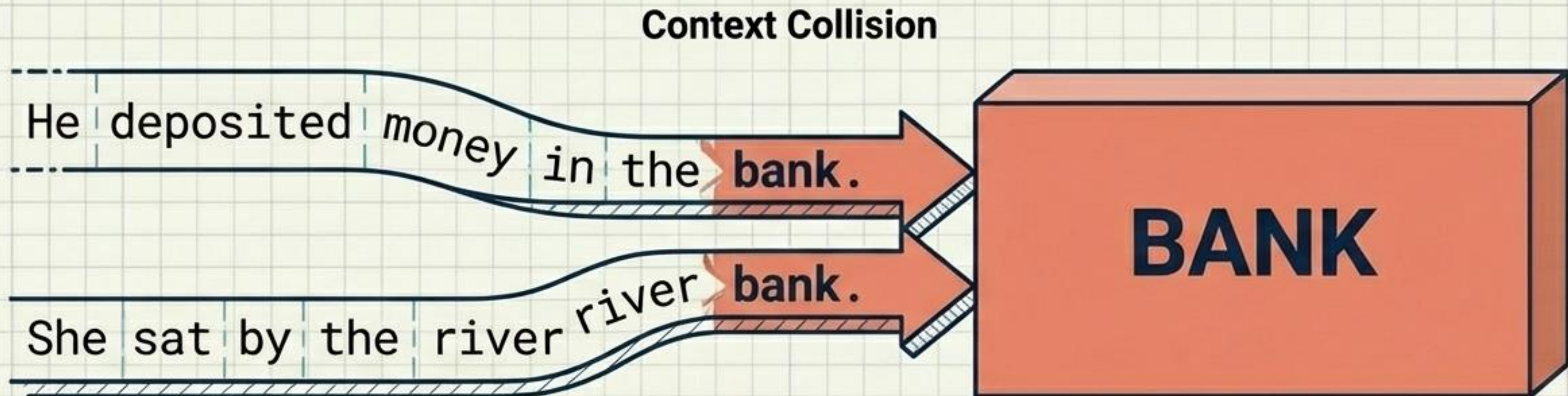


Meaning in language resides not in isolated symbols,
but in the internal and external patterns that connect them.

fastText
fasttext.cc

The Static Limitation

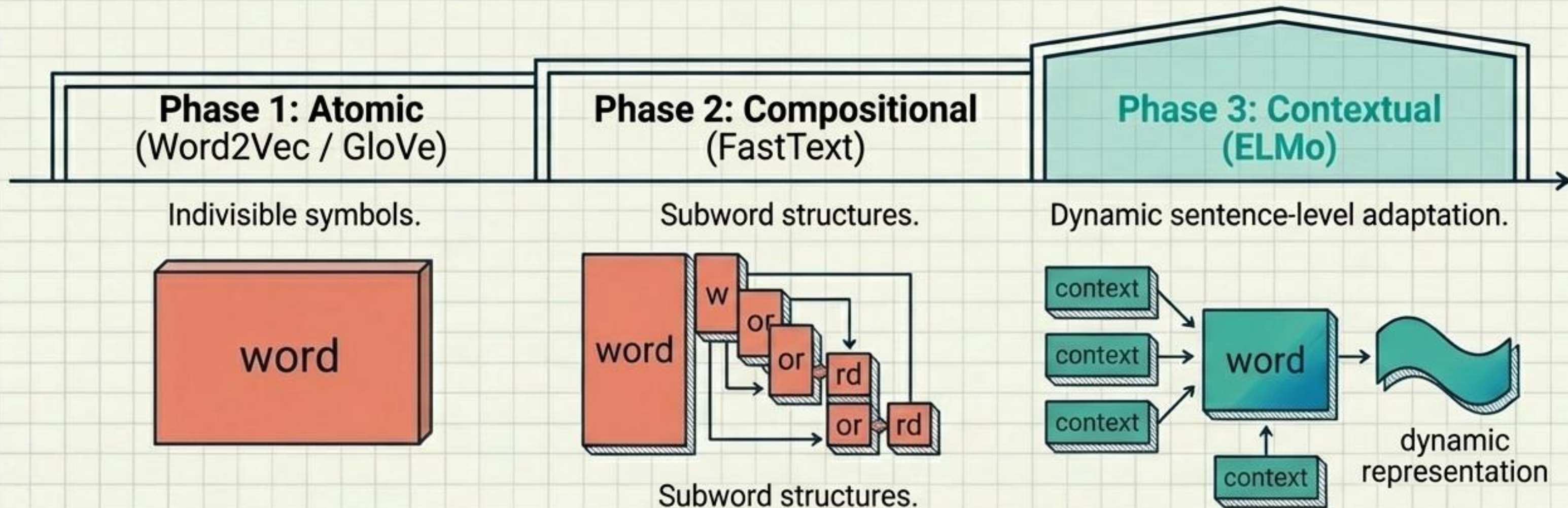
Despite its structural brilliance, FastText assigns a single, frozen vector to every word, regardless of the surrounding sentence.



FastText enriches representation through morphology, but **meaning remains static**. It cannot dynamically adapt to context. The financial institution and the riverside share the exact same mathematical coordinates.

The Next Horizon: Contextual Embeddings

Moving from static composition to dynamic interpretation.



The next evolution in natural language processing abandons fixed vectors entirely. Meaning is no longer static—it becomes a **fluid function of context**.

Support Independent Research: Fueling Discovery

The Financial Barriers to Research



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